AI Deep Drawability

June 2026 Project Update

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Summary

This report explains the current state of the AI Deep Drawability project and outlines future steps.

1 The Data

Currently we have data on 8 unique sheets. Three of these are “good” samples (4, 6, 7) meaning that the material had expected failure rates. The remaining five sheets (1, 3, 5, 2123, 8086) are “bad” meaning that they have abnormal failure rates, indicating issues with the raw material. Currently, each of these sheets is imaged with an Olympus LEXT OLS5100 using the confocal laser. Each sheets has between 12,000 - 16,000 intensity images totaling 118,848 individual samples. Each of these images is represented in greyscale as a 256 $\times$ 256 grid.

The first goal of this week’s work was getting a better understanding of the data we are working with. Figure 1 shows violin and box plots of the mean intensities per sample in each sheet. Comparing the good to bad samples it seems like the good generally have slightly higher mean intensity though with a sample size of three this could come down to random variance. Additionally, sheet 6 immediately stands out as it not only has a much larger range of mean intensities than any of the other sheets but also appears to show bimodal behavior.

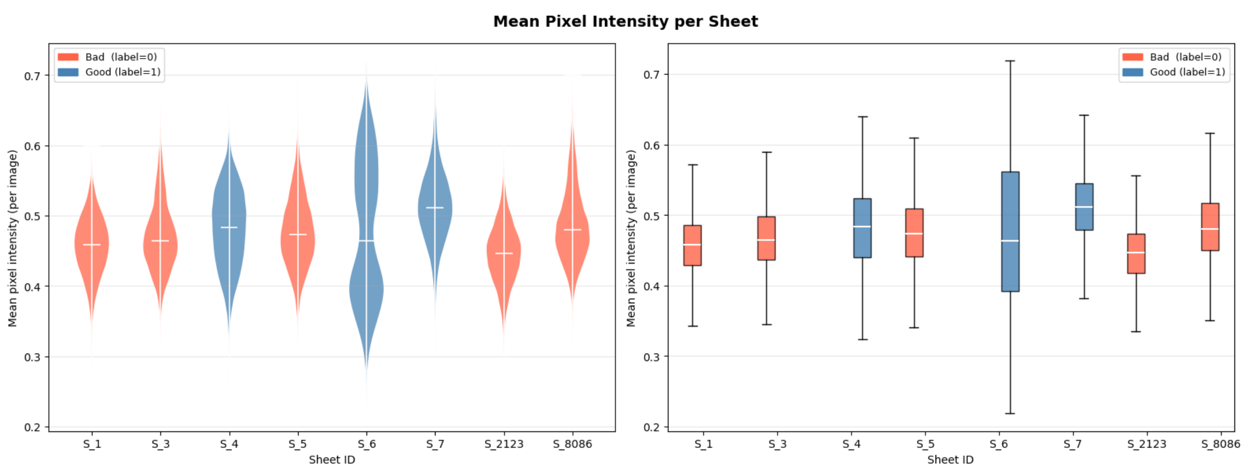

Figure 1: Mean intensity violin and boxplots

Investigating this further in Figure 2, we can see this bimodal behavior even more clearly. This indicates there may be an issue with the imaging for sheet 6 or sheet 6 may truly have some different underlying characteristics. Ignoring sheet 6 for now, we see each sheet generally follows a clean gaussian distribution with the good sheet’s distributions generally centered slightly higher than the bad ones.

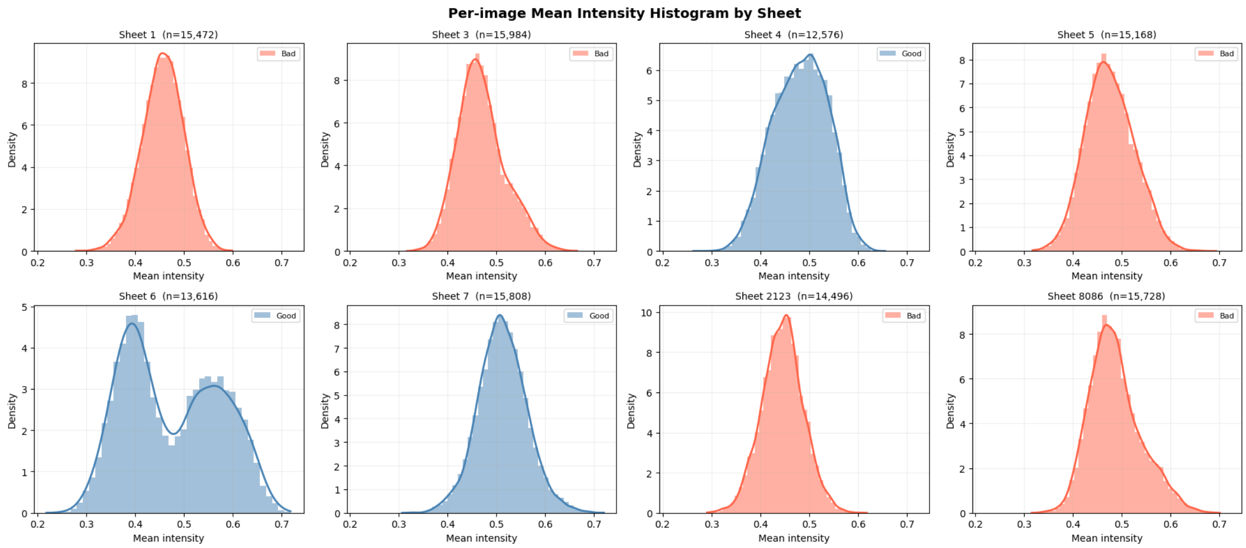

Figure 2: Mean intensity distributions by sheet

Figure 3 explores the standard deviation of our samples. Again, we can see that sheet 6 looks anomalous but now sheet 7 also stands out. Sheet 7 has considerably lower standard deviation than any of the other samples with a mean of just 0.125 compared to the others which are upwards of 0.175.

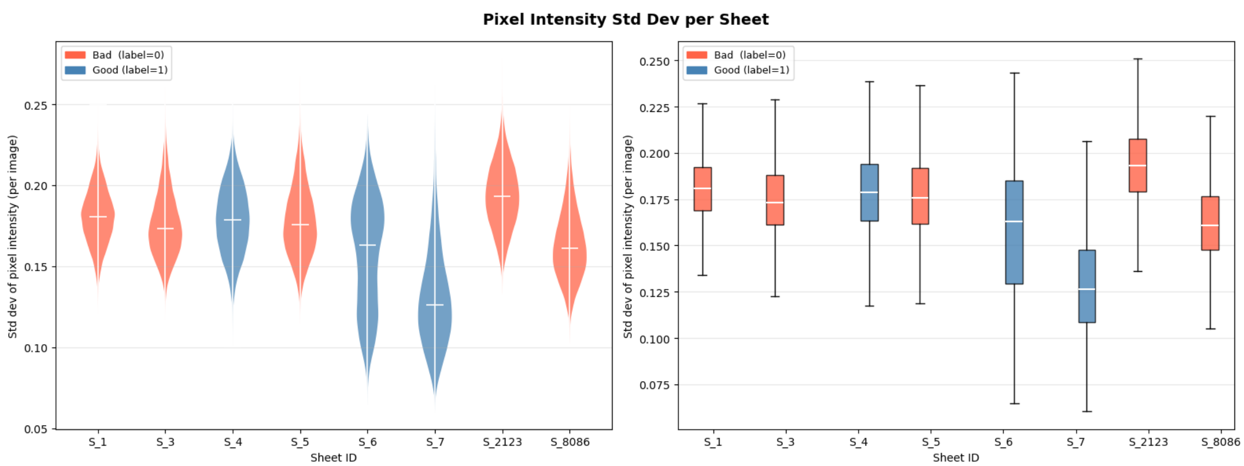

Figure 3: Standard deviation of pixel intensity violin and boxplots

2 Inherited Model

The current model passes the raw image data through Meta's pretrained ConvNeXtV2 model and stores the extracted features as 768 dimensional embeddings. We then take one good and bad sheet out completely as a holdout set to test real world performance. The remaining data is split 80/20 into train and validation sets. Our accuracy looks promising on the validation set at around 0.918 but when tested on the current unseen holdout set (sheet 4, and 8086) this drops significantly to only 0.55. This indicates that our model is likely learning specific sheet to sheet characteristics rather than the underlying patterns of what truly makes a sample "good" or "bad".

3 Updates

After getting a grasp for the data and producing some visuals, the next step was to test on every possible holdout set. Figure 4 displays the accuracy result for each possible holdout combination. We can see that the test accuracies vary significantly depending on the specific sheets we holdout. Sheet 2123 stands out as consistently having the lowest accuracies while sheets 6 and 7 generally show the highest unless paired with 2123. Looking back at Figure 3, we can see that sheet 2123 had the highest standard deviation so this likely played a role in making the model perform worse. Sheet 7 had the lowest standard deviation so this may be attributed to the better performance. Sheet 6 however makes less sense. It alone displays the unique bimodal behavior discussed earlier, so it's relatively high accuracy when used as a holdout seems counter-intuitive.

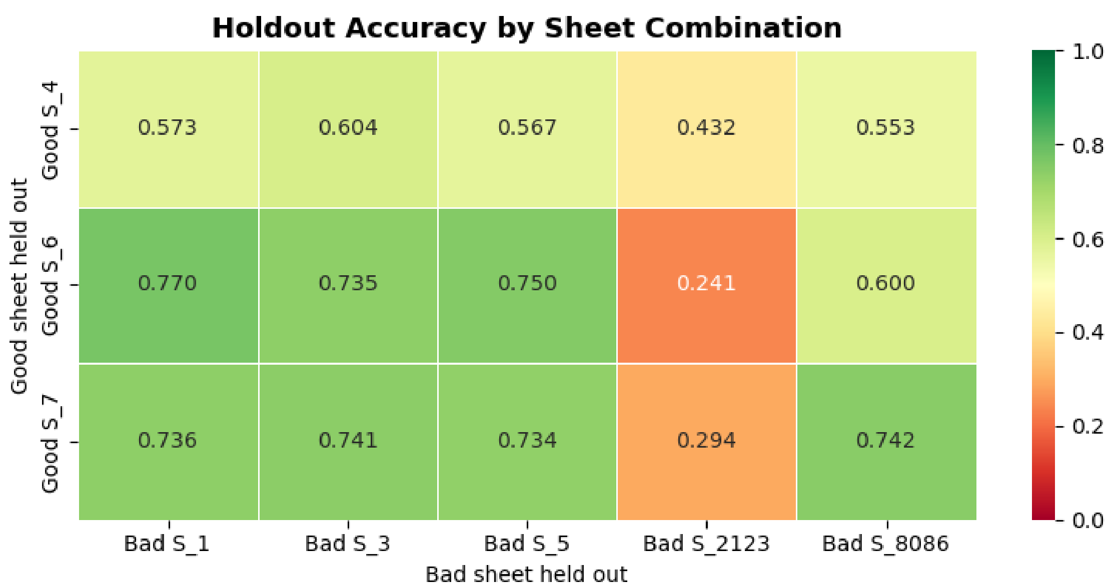

Figure 4: Heatmap of accuracies for all holdout combinations

Moving beyond accuracy and using precision and recall allows us to see the tradeoff between false positives and false negatives. In Figure 5 we see that our model consistently has higher recall than precision. This means it is catching the truly bad samples the majority of the time and has relatively few false negatives. However, this is at the cost of precision meaning that our model often mislabels good samples as bad. In a real world scenario, catching truly bad sheets is the main goal and having some false positives mixed in will is a worthwhile sacrifice to avoid missing bad sheets. So this slight imbalance alone is not a big issue. What is a big issue is the results we see for sheet 2123. We now see why the accuracies for sheet 2123 were so much worse. When used as a holdout, our model classifies every sample in sheet 2123 as good despite it being bad. This leads to a recall and precision of 0. We would expect this sort of issue with extreme class imbalance but since we have a roughly 2:1 ratio of bad to good samples this result is very strange and points to sheet 2123 either following a very different structure or some issue in our model.

I tried adjusting the model architecture to 3 layers with 512, 128, 32 nodes respectively instead of the current single layer of 64 nodes but this did not net any performance improvements. I tried playing with additional hyperparameters but as was previously found, they seemingly don't have a large impact on performance here.

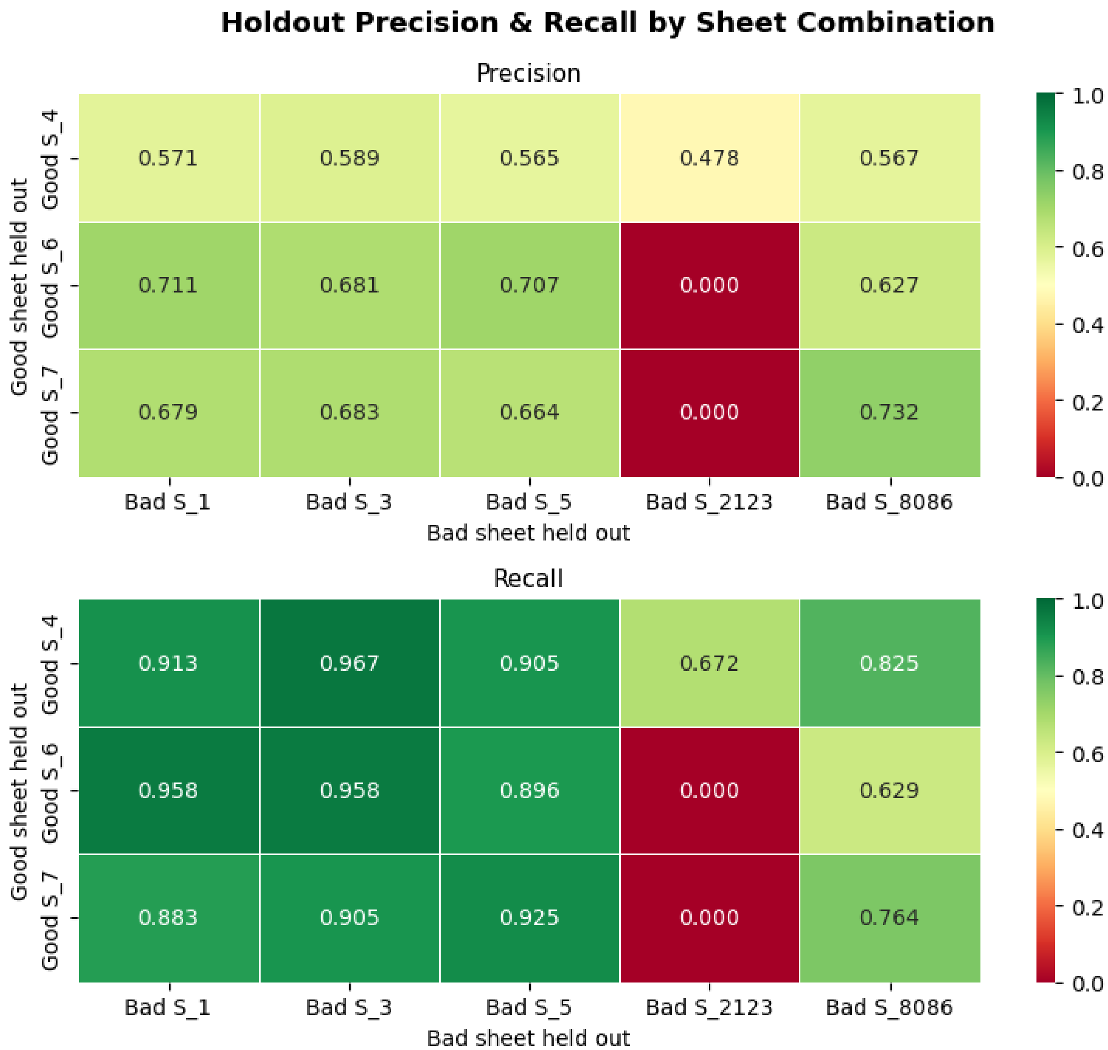

Figure 5: Heatmaps of precision and recall for all holdout combinations

Figure 6 shows an additional exploration of the data looking at performance on the good versus bad sheets. We can again see that our model is much more accurate on predicting bad sheets than good which we already found earlier. If we plot our data distribution side by side with our per-class accuracies there we see a strong similarity in our accuracy on each class and the amount of samples for that class in the training data. I think the next step to address this will be testing a weighted loss function to address the imbalance.

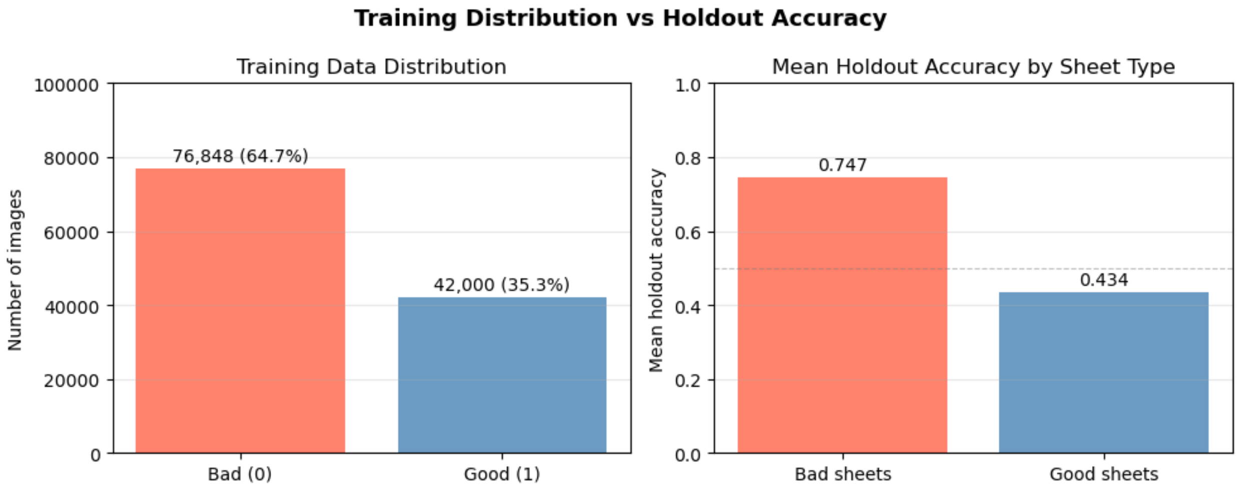

Figure 6: Data distribution and class accuracy

4 Next Steps

First off, a deeper dive into the strange results for some of the sheets is needed. More experimentation with the model settings would also definitely be good as our current results show a strong imbalance between accuracy on the good versus bad sheets. Additionally, I think tuning the ConvNeXtV2 model on our data and adjusting weights rather than just feeding the data through might be valuable although runtime may become an issue. I would also like to test Meta's new DINOv3 pretrained model to see if that leads to any improvements. Then, the next thing to explore will be contrastive learning using a different type of model that will hopefully be able to extract the more meaningful features that separate good from bad. One additional interesting exploration would be to train a classifier to predict specific sheets because I suspect that our current model is just learning per sheet patterns and memorizing which sheets are good and bad rather than actually learning what good and bad look like. Ultimately, I think what is needed is data on more sheets to get over the noise of sample to sample variation and learn what truly separates the good sheets from the bad.